# Thermodynamic description of the Mg–Si system: a synthetic benchmark document

*ORCHID synthetic benchmark (generated; no human authors)*

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**Abstract**

This synthetic document describes the Mg–Si binary system as calculated with pycalphad from the SGTE dataset MgSi-12Sch.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → Mg2Si + (Si), lies at 939.3 °C.

## 1. Introduction

The Mg–Si system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound, simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of M. Schick, B. Hallstedt, A. Glensk, B. Grabowski, T. Hickel, M. Hampl, J. Groebner, J. Neugebauer, R. Schmid-Fetzer, Calphad, 37, 77-86(2012). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

The melting point of Mg reproduced by the dataset is 652.1 °C. Pure Si melts at 1413.8 °C.

## 2. Phase diagram and invariant reactions

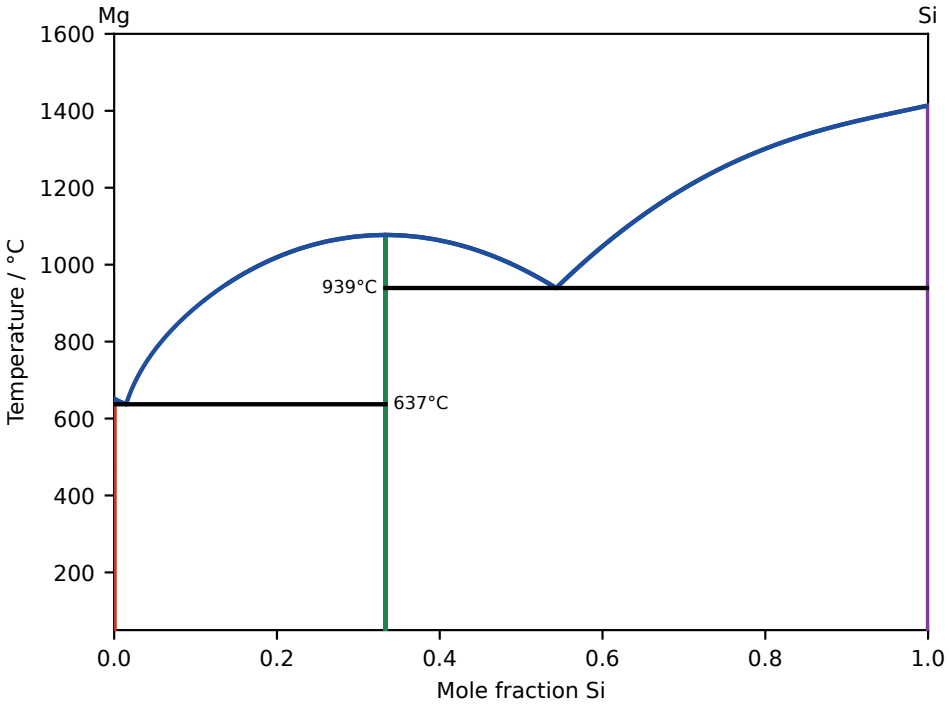



Fig. 1. Calculated Mg–Si phase diagram.

At 939.3 °C the eutectic L → Mg2Si + (Si) takes place; the L phase then contains 54.3 at.%. The eutectic reaction L → (Mg) + Mg2Si is calculated at 637.1 °C, with L at 1.5 at.% Si. Mg2Si melts congruently at 1077.4 °C and 33.3 at.% Si.

Table 1. Invariant reactions and congruent points, calculated (Mg–Si).

| Reaction | Type | T (°C) | x(Si), 1 | x(Si), 2 | x(Si), 3 |
|---|---|---|---|---|---|
| L → Mg2Si | congruent melting[a] | 1077.4 | 0.333 | 0.333 | – |
| L → Mg2Si + (Si) | eutectic | 939.3 | 0.543 | 0.333 | 1.000 |
| L → (Mg) + Mg2Si | eutectic | 637.1 | 0.015 | 0.000 | 0.333 |

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.
[a] Two-phase equilibrium at a temperature extremum; both phases have the same composition.

## 3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Mg–Si).

| Phase | Pearson | SB | Prototype | T min (K) | T max (K) | Range (x(Si)) | T (K) |
|---|---|---|---|---|---|---|---|
| (Mg) | hP2 | A3 | Mg | < 300[a] | 923.0 | 0.000 | 910 |
| Mg2Si | cF12 | C1 | CaF2 | < 300[a] | 1350.5[b] | 0.333 | 300 |
| (Si) | cF8 | A4 | C (diamond) | < 300[a] | 1687.0 | 1.000 | 300 |

Structure labels follow the TDB phase names (– where the name does not identify a structure).
Stability range: temperatures between which the phase is stable in some part of the diagram.
[a] Stable at the lowest calculated temperature.
[b] Congruent point (melting or transformation).

According to Table 2, Mg2Si remains stable up to 1077.4 °C (congruent melting).

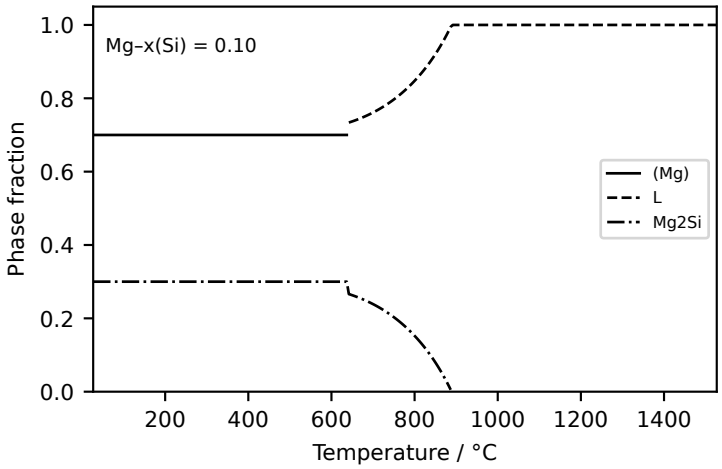



Fig. 2. Calculated equilibrium phase fractions versus temperature for Mg–x(Si) = 0.10.

## 4. Thermodynamic properties

At 1750 K the integral mixing enthalpy of the liquid has its minimum, −16345 J/mol, at 48.0 at.% Si. The calculated enthalpy of formation of Mg2Si at 298.15 K is −21.05 kJ/mol per mole of atoms. Its heat capacity at room temperature is 22.92 J/(mol K). For the equiatomic liquid at 1750 K, the Gibbs energy of mixing is −19.16 kJ/mol.

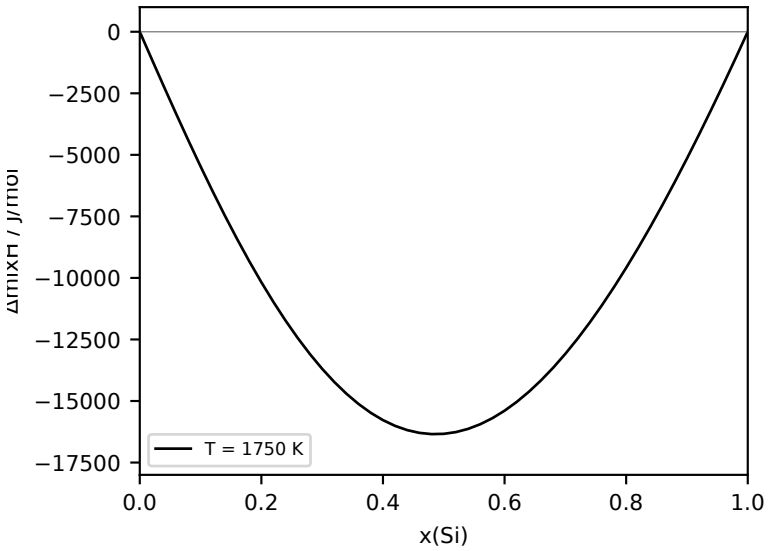



Fig. 3. Calculated mixing enthalpy of the liquid at 1750 K.

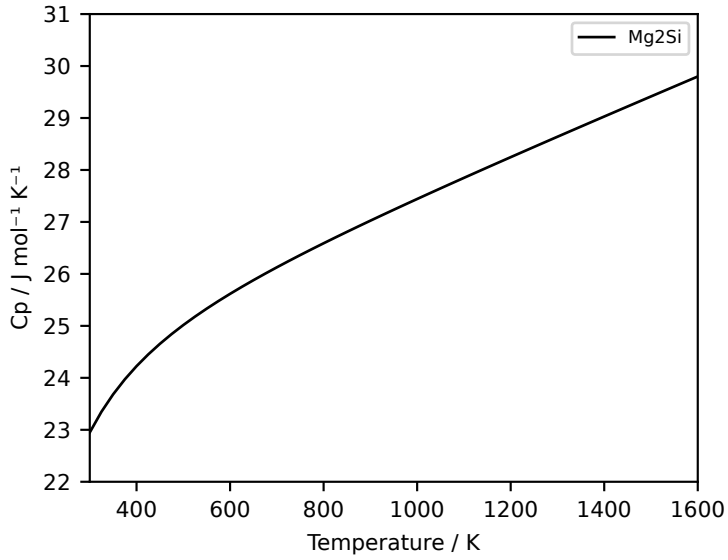



Fig. 4. Calculated heat capacity of Mg2Si.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Mg–Si).

| Phase | at.% Si | ΔfH(298.15 K) / kJ mol⁻¹ (f.u.) | ΔfS / J mol⁻¹ K⁻¹ (f.u.) | Cp(298.15 K) / J mol⁻¹ K⁻¹ (f.u.) |
|---|---|---|---|---|
| Mg2Si | 33.3 | −63.15 ± 2.0 | −8.12 | 68.75 |

Reference: stable elements (SER) at 298.15 K; values per mole of formula units (formula unit as written in the Phase column). ± values are synthetic illustration uncertainties, not data.

Table 4. Integral and partial mixing properties of the liquid at 1750 K (Mg–Si).

| at.% Si | ΔmixH / J mol⁻¹ | ΔmixG / J mol⁻¹ | a(Mg) | a(Si) |
|---|---|---|---|---|
| 10 | −5470 | −8120 | 0.8764 | 0.0124 |
| 20 | −10175 | −13291 | 0.7196 | 0.0387 |
| 30 | −13693 | −16738 | 0.5493 | 0.0874 |
| 40 | −15777 | −18673 | 0.3858 | 0.1687 |
| 50 | −16334 | −19157 | 0.2476 | 0.2902 |
| 60 | −15394 | −18218 | 0.1453 | 0.4489 |
| 70 | −13082 | −15889 | 0.0789 | 0.6242 |
| 80 | −9590 | −12215 | 0.0399 | 0.7834 |
| 90 | −5149 | −7204 | 0.0175 | 0.9043 |

Reference states: pure liquid Mg and Si at 1750 K; per mole of atoms.

## 5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

## References


[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] M. Schick, B. Hallstedt, A. Glensk, B. Grabowski, T. Hickel, M. Hampl, J. Groebner, J. Neugebauer, R. Schmid-Fetzer, Calphad, 37, 77-86(2012).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.